Project Spark — Level 13 Creation Process

Level: `level_13` — **Loaded Voltage Divider Challenge**

Unity: existing Project Spark single-scene loader, shelf, circuit solver and universal workstation bridge

Import file: `Level_13_LoadedVoltageDivider_Import.txt`

1. What this package does

The TXT replaces the data inside the **original** `Level_13.asset` (same GUID, path, JSON field structure and target count). It **does not create or modify a Unity prefab**, place resistors, connect cables or assign bindings. Perform the following steps in the editor.

2. Circuit to build

```
Battery +5 V -- SW1 -- R1 220 ohm --+--- R2 330 ohm ---+--- Battery -
                                    |                   |
                                    +--- R3 470 ohm ---+
                                    |                   |
                                    LCD A               LCD B
                                   (Vout)              (GND)
```

R1 is **in series with the entire parallel load** (R2 || R3). Do not place R1 parallel with the other resistors as in Level 12. The LCD probes go across the **load**: LCD A to Vout and LCD B to negative/GND. Do not put the LCD in series.

3. Calculated expected results (ideal 5 V supply)

Measurement	Correct 330 ohm R2	Wrong 150 ohm R2	
Effective parallel load (R2		R3)	193.875
LCD Vout	2.342 V	1.704 V	
R1 / supply current	12.081 mA	14.983 mA	
R2 current	7.098 mA	11.358 mA	
R3 current	4.983 mA	3.625 mA	

Small deviations from wire/switch resistance are expected. **Inspect the SparkResistor's actual** `Resistance` **field** for each physical component; do not trust the label. Level 11 previously failed when a resistor named 470 ohm had 330 ohm behavior.

4. Import the Level 13 definition

1. Back up the existing `Level_13.asset` and `Level_13_Layout` prefab.

2. Exit Play Mode. Select the **original** `Assets/My_Assets/_Project_Spark/Prefabs/Levels/Level_13_Layout/Level_13.asset`.

3. In your editable TXT importer, import `Level_13_LoadedVoltageDivider_Import.txt` into the selected asset.

4. Confirm `levelId=level_13`, display name **Loaded Voltage Divider Challenge**, four targets and 5 V nominal source.

5. Save assets. The import alone cannot make the topology correct.

5. Create the Level 13 prefab

1. Duplicate your existing working level-layout prefab as `Level_13_Layout`, or edit your already-created original Level 13 prefab. Keep its `SparkLevelLayout`, **one primary** `SparkLevelSceneBindings`, battery, switch, LCD and level-workstation infrastructure.

2. Assign `Level_13.asset` to the prefab's `SparkLevelLayout.Level Definition`.

3. Position a fixed R1 **220 ohm** resistor between SW1 output and Vout. Configure its component resistance to 220 ohm.

4. Position a fixed R3 **470 ohm** resistor between Vout and negative/GND. Confirm its actual resistance is 470 ohm.

5. Provide **one replaceable R2 workstation slot** in the R2 parallel branch; its `SparkShelfSlot.Kind` must be **Workstation**, not Storage. Make sure it is registered with the shelf when the layout loads.

6. Use a 150 ohm resistor as the initial *incorrect* installed R2 if you want the challenge to start assembled; otherwise let the player first install it. The correct choice is a 330 ohm shelf resistor.

7. If the prefab came from an older level, remove/retire its obsolete level-specific disposable bindings without deleting permanent battery/switch/LCD references.

6. Connect the wires exactly

#	From	To
1	Battery_positive	SwitchInput
2	SwitchOutput	fixed Resistor_220_Input (R1)
3	fixed Resistor_220_Output (R1)	R2 workstation **input** (Vout)
4	fixed Resistor_220_Output (R1)	fixed Resistor_470_Input (R3)
5	R2 workstation **output**	Battery_negative
6	fixed Resistor_470_Output (R3)	Battery_negative
7	LCD_Terminal_A	Vout at R2 input / R3 input
8	LCD_Terminal_B	negative/GND at R2 output / R3 output

The terminal labels in the table are **logical references**; use the real objects, and ensure the physical graph connections exist. The circuit solver should show R1 current equal to the **sum** of R2 and R3 branch currents.

7. Configure authoritative scene bindings

Primary fixed bindings on `Level_13_Layout`**:** `Battery_positive`, `Battery_negative`, `PowerSupply_01`, `Resistor_220_Ω`, `Resistor_220_Input`, `Resistor_220_Output`, `Resistor_470_Ω`, `Resistor_470_Input`, `Resistor_470_Output`. Keep any existing permanent switch/LCD binding IDs required by your systems. Create or rename fixed IDs only to match actual objects; Unity names alone do not establish binding IDs.

Declared runtime IDs: `R2_Component`, `R2_Input`, `R2_Output`. These are logical IDs and **must not point to a specific shelf resistor prefab**.

On the **original Level 13 layout prefab root**, configure exactly one `SparkLevelWorkstationBindingBridge` rule:

Bridge field	Value
Slot Id	The actual active Level 13 R2 **workstation** SparkShelfSlot.SlotId (for example Register_Slot_2 only if this is actually the one)
Component Binding Id	R2_Component
Input Terminal Binding Id	R2_Input
Output Terminal Binding Id	R2_Output
Electrical Type Name	SparkResistor
Input Terminal Name	_Input
Output Terminal Name	_Output
Exact Terminal Names	unchecked

Click **DECLARE / CHECK RUNTIME IDS IN PRIMARY BINDINGS** in **Edit Mode on the original prefab**, verify three declarations, and save the prefab. Runtime item association then happens automatically. **Do not use storage slot `Placement_01_04` as your workstation slot.**

8. Level target configuration

Target ID	Type	Component / terminals	Pass range
loaded_divider_r1_conducting	ComponentConducting	Resistor_220_Ω	conducting
loaded_divider_r2_current	CurrentRange	R2_Component	0.0068–0.0074 A
loaded_divider_output_voltage	VoltageRange	R2_Input vs R2_Output	2.25–2.45 V
loaded_divider_source_current	CurrentRange	PowerSupply_01	0.0118–0.0124 A

The output voltage target uses `R2_Input` and `R2_Output`, which are the same two nodes as the R3 load. Your evaluator measures absolute terminal-pair voltage. Source current has `Require Conduction=false`, consistent with the tested Level 11 asset; the actual current-range check remains enforced.

9. Test in Unity

■ Open the original Level 13 prefab, validate fixed binding references, save.

■ Verify the Layout Loader is registered to use this prefab for the original Level 13 asset.

■ In Play Mode verify `Level_13_Layout(Clone)` exists and the active level is `level_13`.

■ Verify the R2 bridge log says `source=SLOT_OCCUPANT` and `bound=Resistor_330...` after installing 330 ohm.

■ Inspect resistor actual configured resistances: R1=220, R2=330, R3=470 ohm.

■ Verify R1 in series with the parallel R2/R3 load, not in parallel with them.

■ Verify LCD about 2.342 V, R2 about 7.10 mA, supply about 12.08 mA.

■ Check `LevelGamePlayManager.TargetResults` for all four `true`.

■ Verify current limits on source and no overload/short reported.

■ Save original assets outside Play Mode. Don't save state from a runtime `(Clone)`.

10. Troubleshooting without editing source code

• **`SLOT_MISSING`:** bridge `Slot Id` points to a storage slot or an unregistered workstation; inspect `SparkShelfSlot.Kind`, `SlotId`, registration.

• **`item=EMPTY`:** R2 resistor is not installed in the workstation or shelf occupancy is not settled; inspect `Occupant`, `CurrentSlot`, `IsAtWorkbench`.

• **R2 aliases unresolved:** check original prefab runtime ID declarations, exact spelling and active primary bindings.

• **Wrong voltage/current:** check the real resistance values first, then circuit topology and wire connections. Never force the target thresholds to compensate.

• **3/4 targets:** log target result IDs; inspect the actual source current through the source-current audit rather than assuming the first target's 220 ohm current is source current.

• **No level clone:** verify `CanLoadLevel`, primary binding validation, loader registration and progression access. Do not change the solver.

Scope: This is an asset-data import and a manual build guide; it does not automatically generate a Unity scene or prefab. No C# changes are needed if the existing universal bridge and evaluator work as they did for Levels 11–12.